

BUILDING DYNAMIC FINANCIAL DASHBOARDS USING POWER BI

Training report submitted in partial fulfillment of the requirements for the

Award of the degree of

MASTER OF BUSINESS ADMINISTRATION

Of

BENGALURU CITY UNIVERSITY



Submitted By

Harpriya Behera

[P18W23M015009)

Under the guidance of

Dr R.N. Subba Rao

Principal



ST. FRANCIS COLLEGE

BENGALURU CITY UNIVERSITY

Amritanshu Sinha

UDYOG BHARAT

EXECUTIVE SUMMARY

The MBA course offered by St. Francis College, affiliated with Bengaluru City University, Karnataka, is designed in a manner to provide theoretical as well as practical training. As a part of this, MBA students are required to go through a training program for 30 hours in their 3rd semester.

This program is an integral component of the Business Management course and gives students actual experience in studying how businesses are managed. As a participant in this program, students are able to experience how each department works as a unit, understand processes in businesses, and gain practical experience in a real business environment. It is in this manner that students are not only given theoretical learning but are equipped with practical expertise as well in corporate life.

TABLE OF CONTENT

CHAPTER NO.	TITLES	PAGE NO.
1.	INTRODUCTION 1.1 Introduction to the training 1.2 Importance of the training 1.3 Need of the training	1-4
2.	DESCRIPTION OF THE TRAINING 2.1 Type of training undergone 2.2 The usefulness of the training undergone	5-7
3.	TRAINING OUTCOME 3.1 Knowledge acquired 3.2 Skills mastered 3.3 Proof for operations using skills	8-24
4.	PLACEMENT OPPORTUNITIES 4.1 Placement opportunities in India 4.2 Global Placement opportunities	25-27
5.	LEARNING EXPERIENCE 5.1 Brief on new skills learned 5.2 Challenges faced during training	28-29

CHAPTER – 01

INTRODUCTION

1.1 INTRODUCTION TO THE TRAINING

If you work in finance, you need to review large amounts of data every day. Keeping track of financial statement, working capital, assets, and more can feel overwhelming. However, financial dashboards can make these tasks much easier by putting all of the key metrics you rely in one convenient location. A financial dashboard is a visual tool that summarizes and displays key financial metrics and KPIs (Key Performance Indicators) in an easy-to-understand format, providing a quick overview of an organization's financial health. For example, you may need to analyze accounts receivable and accounts payable reports. At the same time, you may also need to track balance sheet figures like assets, liabilities, and equity.

Under academic courses, we conducted an offline session of 30 hours in Building Dynamic Financial Dashboard using Power BI in our college campus in association with Udyog Bharat. This training provided hands-on experience in methods used to financial dashboard and working with Power BI, an increasingly popular business intelligence tool to visualize and analyze data.

Power BI is required in corporations because it helps individuals analyze vast amounts of data, spot trends, and develop forecasted insights. From this training session, we learned to apply financial techniques to previous revenues, craft dynamic reports, and build interactive dashboards to guide decisions.

The training comprised core concepts in financial analysis, trends, seasonality, and fluctuations in revenues. We studied methods employed to obtain improved prediction results. Sessions comprised theory sessions and practice sessions where we applied prediction models in Power BI.

Offline training in our college featured interactive sessions, live case studies, and peer collaboration to aid in learning. Sessions featured academic and sector-related knowledge with facilitation by professionals in the industry and instructors.

The training enhanced our analytical and problem-solving competencies to better support management students in various professional careers. Acquiring knowledge in revenue

forecasting can support finance, business analytics, and strategic management. This is transferable to academic projects and future careers.

After this training. We achieved an effective working experience in predicting revenues with Power BI, providing hands-on experience in finance-related data analysis to aid in making informed decisions. This report is an overview of our training experience, key areas covered, and capabilities achieved through this training.

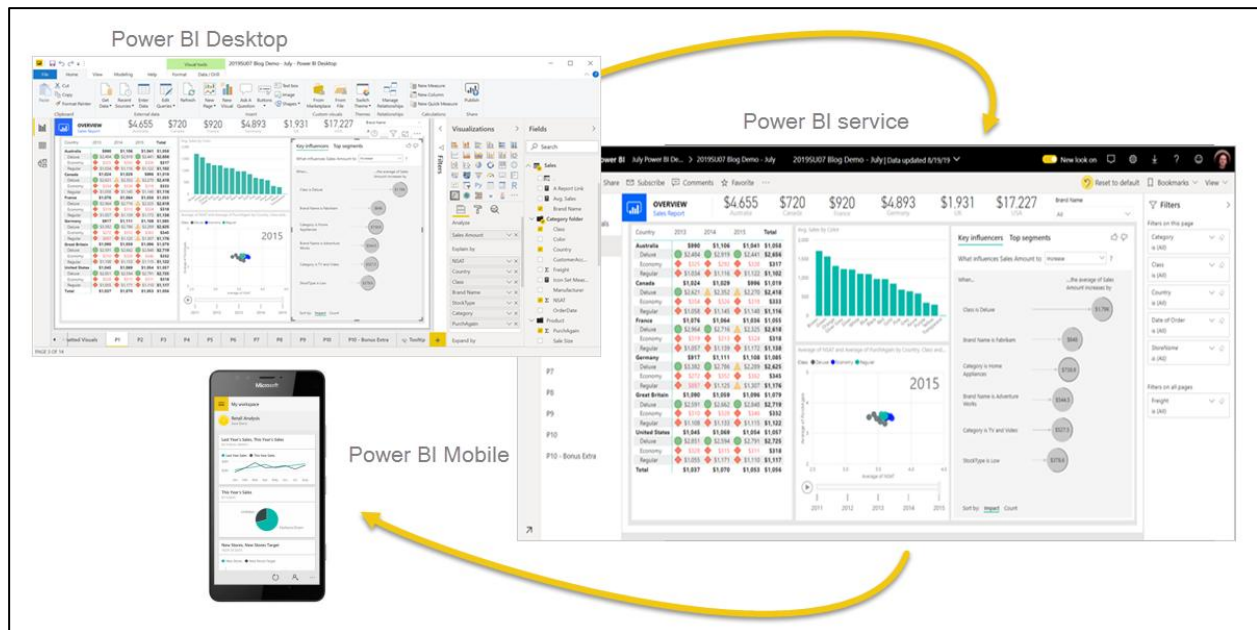
1.1.2 What is Power BI

Power BI is a collection of software services, apps, and connectors that work together to turn your unrelated sources of data into coherent, visually immersive, and interactive insights. Your data might be an Excel spreadsheet, or a collection of cloud-based and on-premises hybrid data warehouses. Power BI lets you easily connect to your data sources, visualize and discover what's important, and share that with anyone or everyone you want.

1.1.3 The Parts of Power BI

Power BI consist of several elements that all work together, starting with these three basics:

- A Windows desktop application called Power BI Desktop.
- An online software as a service (SaaS) service called the Power BI services.
- Power BI Mobile apps for Windows, ios, and Android Devices.



(Figure 1.1 Power BI Elements)

(Source: <https://learn.microsoft.com/en-us/power-bi/fundamentals/power-bi-overview>)

These three elements-

1.2 IMPORTANCE OF THE TRAINING

A financial dashboard is necessary in order to keep track of performance, recognize trends, and make informed decisions utilizing key financial drivers, particularly if rapid access to information and analysis is vital in strategic planning as well as to operate efficiently.

1.2.1 The Role of Financial Dashboard:

- Financial dashboards show difficult financial information clearly and in easily consumable ways, enabling easier understanding and review of performance for stakeholders.
- Through offering an overall picture of financial information, dashboards enable organizations to make strategic and well-informed decisions.

1.2.2 Power BI Role in Financial Dashboard:

- Power BI enables users to present financial information through different charts, graphs, and tables in an interesting and informative manner, thus providing clarity and accuracy in presenting financial data.
- Power BI is used to create interactive financial dashboards that enable users to drill down into data, analyze trends, and understand financial performance in more detail.

1.2.3 Practical Learning & Hands on Experience:

- Analyze financial information to comprehend trends, seasonality, and fluctuations in revenues.
- Create dynamic financial dashboard revenue reports with interactive dashboards.

1.2.4 Career Advantages in Industry:

- It is advantageous for careers in finance, business analytics, sales, and strategic planning.
- Build experience in handling business data to be able to make trustworthy financial in order to possess superior analytical ability.
- Knowledge about Power BI and financial methods provides an employability advantage and makes graduates job-ready.

1.2.5 Advantages of Offline Training in College:

- Engage in interactive learning through dialogue and instant problem solving.
- Obtain training direction by professionals in the field and faculty members.
- Collaborate with peer- grouped datasets and case studies to collaborate in groups.
- It teaches sector-related competencies to work in data-driven decision making. Training to financial dashboards in Power BI enhances efficiency and stability in finance in work environments.

1.3 NEED OF THE TRAINING

The Financial Dashboard is a valuable tool that can help companies improve their understanding of their financial performance and make better decisions about how to manage their business. Training in Financial Dashboard in Power BI enabled us to analyze trends and prepare reliable financial.

1.3.1 The Importance of Financial Dashboard:

- Time and Cost Savings
- Enhanced Decision-Making
- increased Collaboration and Accountability
- Real-time Access to Financial Data

1.3.2 Issues with Classical Financial Dashboard:

- Designing an intuitive user interface
- Real-time data updates and refresh rates
- Mobile responsiveness and accessibility
- Data integration from multiple sources

1.3.3 Power BI Benefits

- Simplifies data interpretation
- It automates predictive analysis
- Enhances visualization
- Cost Effective

1.3.4 Practical Advantages in Terms of:

- Improved Data Accuracy
- Hands-on experience
- Accessibility

1.3.5 Advantages of Offline Training in College:

- Direct interaction with experts
- Collaborative learning
- Instant resolution of queries

The training has provided us with Power BI functionality to forecast revenues in this data driven economies era

CHAPTER-02

DESCRIPTION OF THE TRAINING

2.1 TYPE OF TRAINING UNDERGONE

Building Dynamic Financial Dashboard was an off-line training program that was undertaken for 30 hours in my college in association with Udyog Bharat. The training was undertaken in order to experience hands-on practice in forecasting financial report, an important technique in business decision making as well as in preparing budget plans.

I developed hands-on experience in visualization and financial dashboard from Power BI training. Power BI is a robust business intelligence tool through which data can be analyzed, trends can be found, and decisions can be made. Training taught me how to implement these tools and techniques on financial revenue in the future based on historical data.

Training was designed to combine theoretical learning in harmony with practical experience to acquire overall enriched experience. Sessions were conducted by industry experts and teaching faculty to provide information required regarding implementation in actual business settings.

Training sessions included:

- **Theoretical Learning-** Gaining fundamental concepts in finance dashboards using Power BI.
- **Practice Hands-on-** Application of Power BI in deploying financial dashboard.
- **Case Studies and Industry Examples-** Applying Financial methods to actual-world datasets.
- **Collaborative Learning-** Encouragement to interact with instructors and fellow learners to clarify doubts and reinforce learning.

The step-by-step tutorial taught me how the techniques of financial dashboards are utilized in business analytics and how Power BI is utilized to influence decisions with insights constructed.

2.2 THE USEFULNESS OF THE TRAINING UNDERGONE

The training was really helpful because it not only provided theoretical knowledge but also hands-on experience in financial projection. Key takeaways and value gained through training are:

2.2.1 Understanding Why Financial Dashboard is Important:

- **Clear Communication** - By putting complex financial information in a visual format, dashboards enable easy comprehension and communication of financial metrics by teams.
- **Shared Understanding** - Financial dashboards provide a common platform for stakeholders to look at and share financial data, enabling better communication and collaboration.
- **Performance Tracking** - Dashboards allow businesses to track their performance against their financial goals and implement necessary adjustments.

2.2.2 Learning to Analyze Financial Dashboard to Predict:

- **Revenue & Expenses** - Track total revenue, cost of goods sold (COGS), operating expenses, and net profit.
- **Cash Flow** - Track cash inflows (receiving money) and cash outflows (paying out money) to ascertain liquidity and financial position.
- **Debt & Equity** - To determine a firm's capital composition, examine the levels of debt, equity, and debt-versus-equity ratios.

2.2.3 Mastering Various Financial Dashboard Models:

- **Revenue:** The total amount of money earned by the company.
- **Net profit:** The difference between revenue and expenses. Total revenue minus total expenses.
- **Cash flow:** The amount of cash moving in and out of the business.

2.2.4 Hands-on- Experience with Power BI:

- **Data Import and Transformation** – Learning to import & prepare data to analyze in Power BI.
- **Using Financial Tools** - Power BI allows you to visualize and analyze financial statements like income statements, balance sheets, and cash flow statements.
- **Building Interactive Dashboards** – Designing dynamic reports to display fixed assets and investment in an effective manner.

2.2.5 Acquiring Analytical and Problem-Solving Skills:

- **Financial Statement Analysis:** Familiarize yourself with key financial statements (income statement, balance sheet, cash flow statement) and learn how to interpret their components.
- **Data Visualization:** Use charts, graphs, and dashboards to visualize data and make it easier to spot investment and fixed assets.
- **Key Performance Indicators (KPIs):** Identify and understand the KPIs relevant to your area of focus (e.g., revenue growth, profitability, customer acquisition cost).

CHAPTER – 03

TRAINING OUTCOME

3.1 KNOWLEDGE ACQUIRED

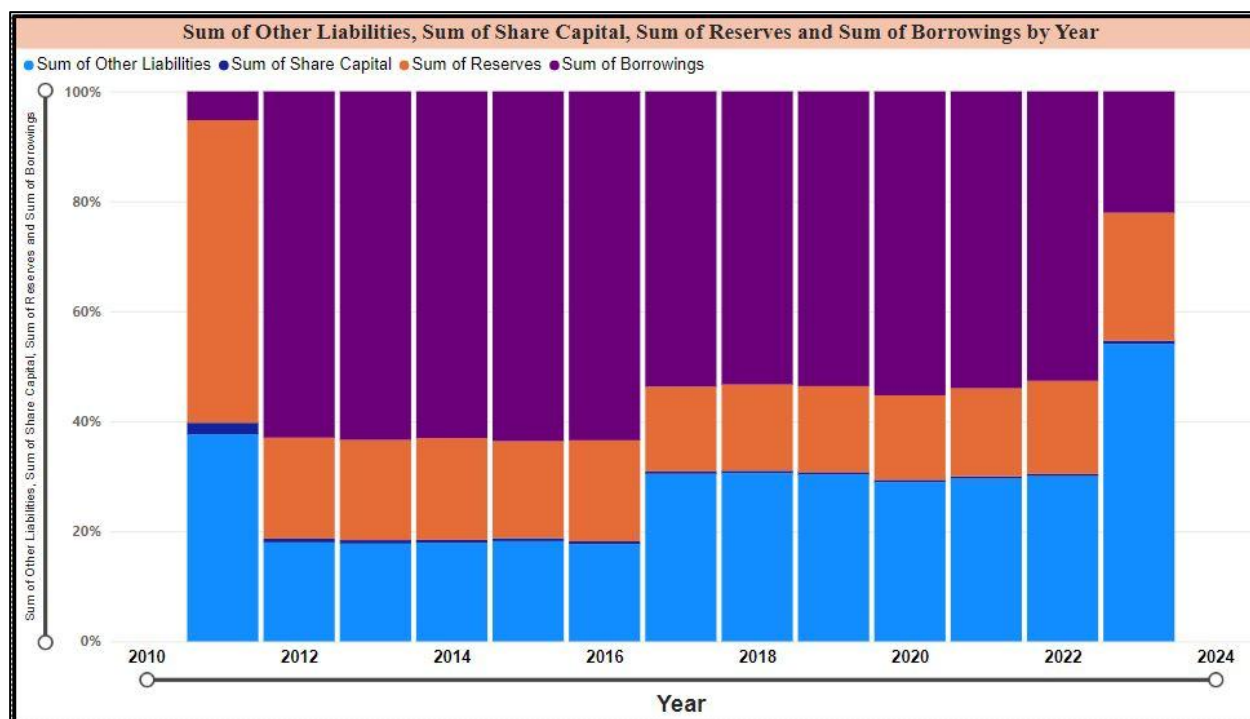
I learned about financial dashboard, visualization, and tools in Power BI 30-hour Training in Power BI: Building Dynamic Financial Dashboard. Training was focused on the use of charts and analytical tools to forecast future financial statement Expenses, Revenue and Profit. Learned to utilize Power BI visuals including bar charts, column charts, pie charts, ribbon charts, tree maps, waterfalls charts, lines charts, and decomposition trees to have in-depth knowledge about finance-related data.

3.1.1 Using Stacked Column Chart for Liabilities Breakdown Over Time

I developed skill in comparing liabilities by categories using Stacked Column Chart.

- X-Axis - quarter(10yrs)
- Y-Axis - Drag Share Capital, Reserves, Borrowings, Other Liabilities
- Stacked Column Chart: The Stacked Column Chart assists in the analysis of the composition of liabilities over time, illustrating how various elements-Share Capital, Reserves, Borrowings, and Other Liabilities are contributing to the total liabilities of the company.
- Every column depicts the total liabilities for a given period of time.
- The different colored sections in each column show how much each liability type contributes.

I was in a position to analyze performance of liabilities breakdown over time in an organized way through Stacked Column Chart.



(Figure 3.1 Stacked Column Chart)

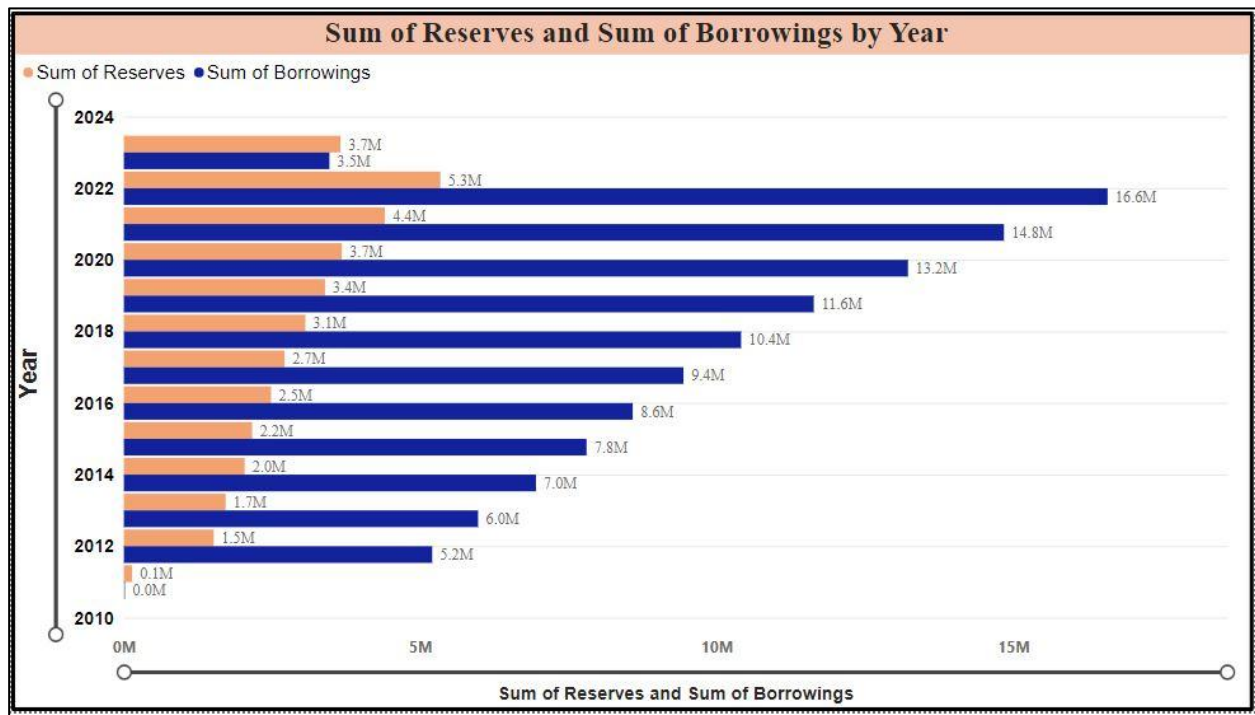
3.1.2 Borrowings vs. Reserves Trend Analysis Using Clustered Bar Chart

The Clustered Bar Chart visually compares the company's Borrowings and Reserves over different financial periods. This helps to understand whether the company is relying more on external debt (borrowings) or internal financial strength (reserves) to support its operations and growth. The chart will now display two bars for each time period, one for Borrowings and one for Reserves.

- X-Axis - quarter(10yrs)
- Y-Axis - Borrowings and Reserves
- This will compare borrowings and reserves side by side.
- If Borrowings bars are consistently higher than Reserves, it means the company is relying more on external debt rather than internal funds.
- If Reserves bars are higher, the company is financially stable and has strong internal funding.

- If Borrowings are increasing while Reserves are stagnant or decreasing, it could indicate financial risk.
- If Reserves are growing along with Borrowings, it means the company is maintaining a good balance.

I was in a position to analyze the performance in business and financial statement in an organized way through Clustered Bar Chart.



(Figure 3.2 Clustered Bar Chart)

3.1.3 Total Assets Performance Using Line Chart

A Line Chart is employed to monitor changes in a metric over time. Here, it is employed to illustrate how the total assets of the company have evolved across several financial quarters. This analysis is critical in understanding financial growth, stability, and possible risks in asset management.

- X-Axis - quarter(10yrs)
- Y-Axis - Total Assets

- If the line is consistently rising, the company is experiencing asset growth, which is a positive sign of financial stability.
- A decline in total assets may indicate financial distress, asset sales, or economic downturns affecting the company.
- Plot Total Assets in column.

Line Chart provided an accurate presentation of Total Assets across over time.

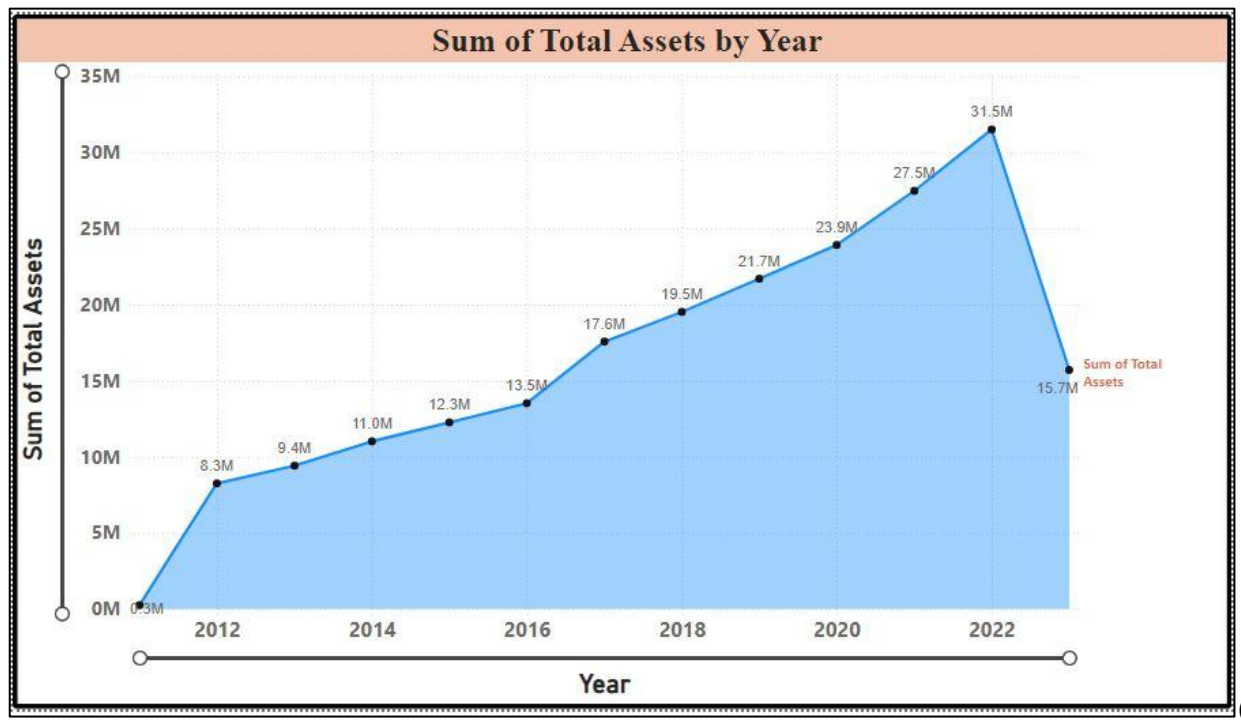


Figure 3.3 Line Chart)

3.1.4 Waterfall Chart: Sum of Share Capital by Year

This Waterfall Chart visually represents the yearly changes in share capital over time. It shows how the company's share capital has grown incrementally from 2011 to 2023, with a final total at the end.

- Category - quarter(10yrs)
- Values - Total Assets
- The waterfall chart will show year-over-year changes in assets.

- Each bar represents the additional share capital added each year.
- The dark blue bars indicate an increase in share capital, showing steady financial growth.
- There are no red (decrease) bars, meaning there were no reductions in share capital during this period.

The final total bar in light blue represents the cumulative sum of all increases, reaching 1.02M in 2023.

The waterfall came in especially handy to comprehend sum of share capital.



(Figure 3.4 Waterfall Chart)

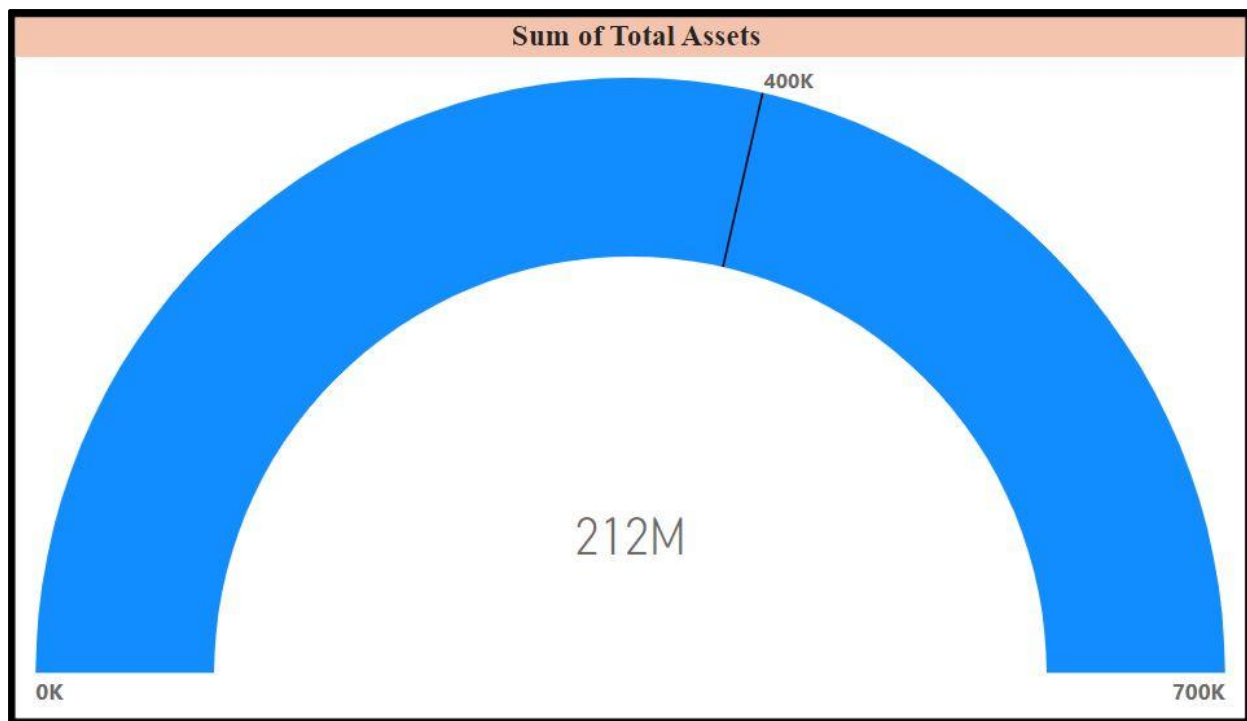
3.1.5 Gauge Chart: Current Total Assets vs. Target

A Gauge Chart is a powerful visual tool that helps track progress toward a specific target. This representation is frequently applied in financial reporting, tracking of performance, and monitoring of goals because it presents a clear and concise indication of how close the firm is to meeting its financial objectives.

- Value - Total Assets

- Set a target manually (e.g., 400000).
- Adjust the gauge range from minimum to maximum values.
- If total assets will never go below 0, set this to 0.
- If assets can vary widely, set it to their lowest ever value for improved visualization.
- Select a figure slightly above the goal (e.g., 700,000) to ensure a clear vision of progress.
- This makes the movement of the needle easy to see and stops the gauge looking full too early.

I was able to determine a Gauge Chart and simple yet effective way to track financial progress



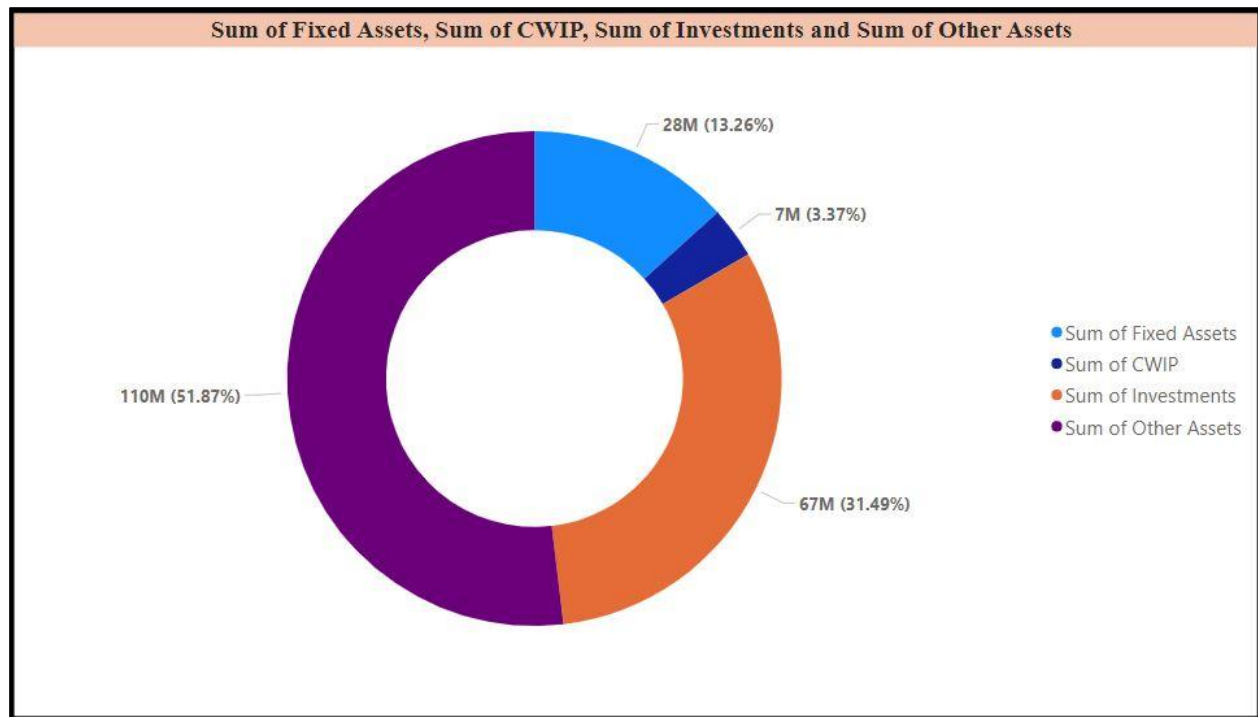
(Figure 3.5 Gauge Chart)

3.1.6 Donut Chart: Breakdown of Fixed Assets, CWIP, Investments & Other Assets

A Donut Chart is a good way of representing the composition of total assets by dividing them into different categories. It facilitates decision-making by enabling the decision-makers to understand the assets' distribution, which assists in deciding in which areas the company's capital is being spent.

Here, we are displaying Fixed Assets, CWIP (Capital Work in Progress), Investments, and Other Assets as different slices of the pie, relating to their proportion with respect to total assets.

- Values - Fixed Assets, CWIP, Investments, Other Assets.



(Figure 3.6 Donut Chart)

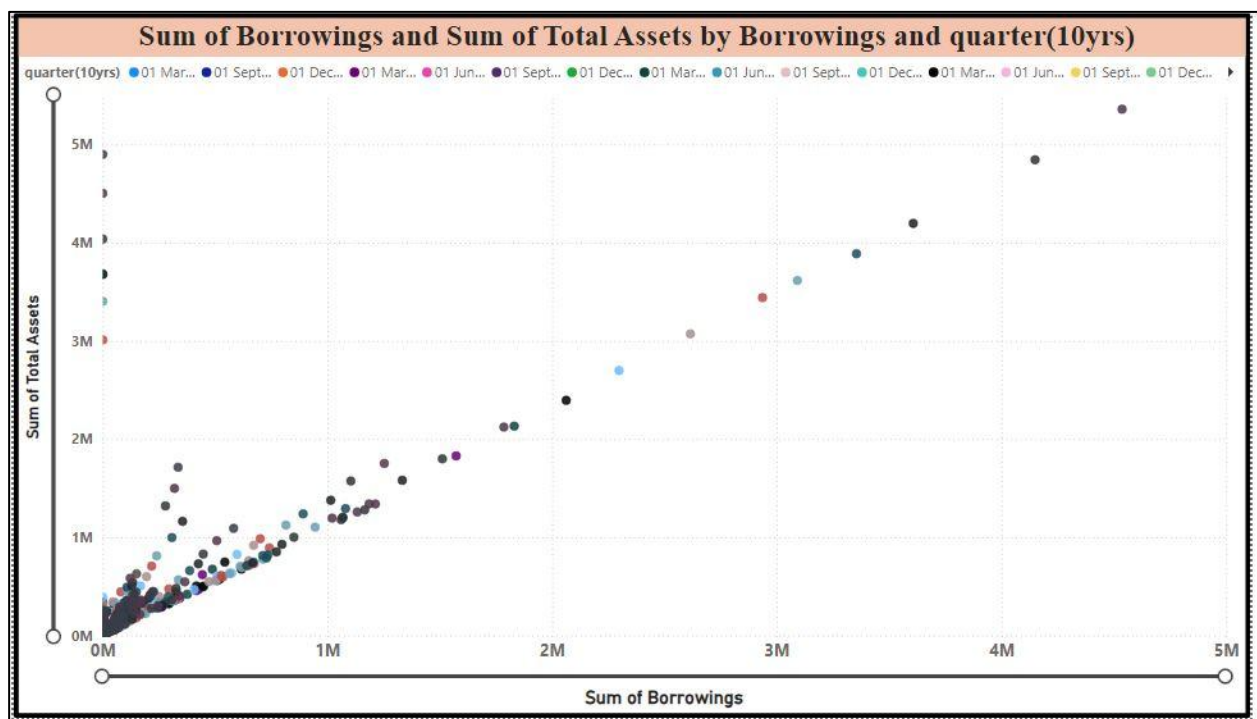
3.1.7 Using Scatter Plot for Analysis of Borrowings Vs. Total Assets

A Scatter Plot is used to visualize the relationship between two numerical variables. In this case, we are comparing Borrowings (X-Axis) and Total Assets (Y-Axis) to determine whether there is a correlation between borrowing levels and asset growth.

By adding quarter(10yrs) as the legend, we can color the points by year to analyze trends over time.

- X-Axis - Borrowings (This represents the amount of borrowed funds).
- Y-Axis - Total Assets (This represents the company's total asset value).
- Legend - quarter(10yrs) (This will color each data point based on the respective year).

- Color Gradient (Year Scale) - Points are colored based on the year to show how trends evolve over time.
- It Determines whether borrowing helps asset growth.
- It Helps in risk assessment, if high borrowing doesn't lead to asset growth, it may indicate inefficient debt management.
- Provides decision-making support for financial strategy whether the company should increase or reduce borrowing.
- Tracks changes over time by using the legend to group data by year.



(Figure 3.7 Scatter Plot)

3.1.8 Financial Overview: KPI Cards for Assets, Borrowings & Reserves

KPI (Key Performance Indicator) Cards provide a quick snapshot of key financial metrics, allowing businesses to monitor important values at a glance.

- First Card - Total Assets to display the company's total assets.
- Second Card - Borrowings to track the total amount of borrowed funds.
- Third Card - Reserves to show the total reserves available.

Total Assets

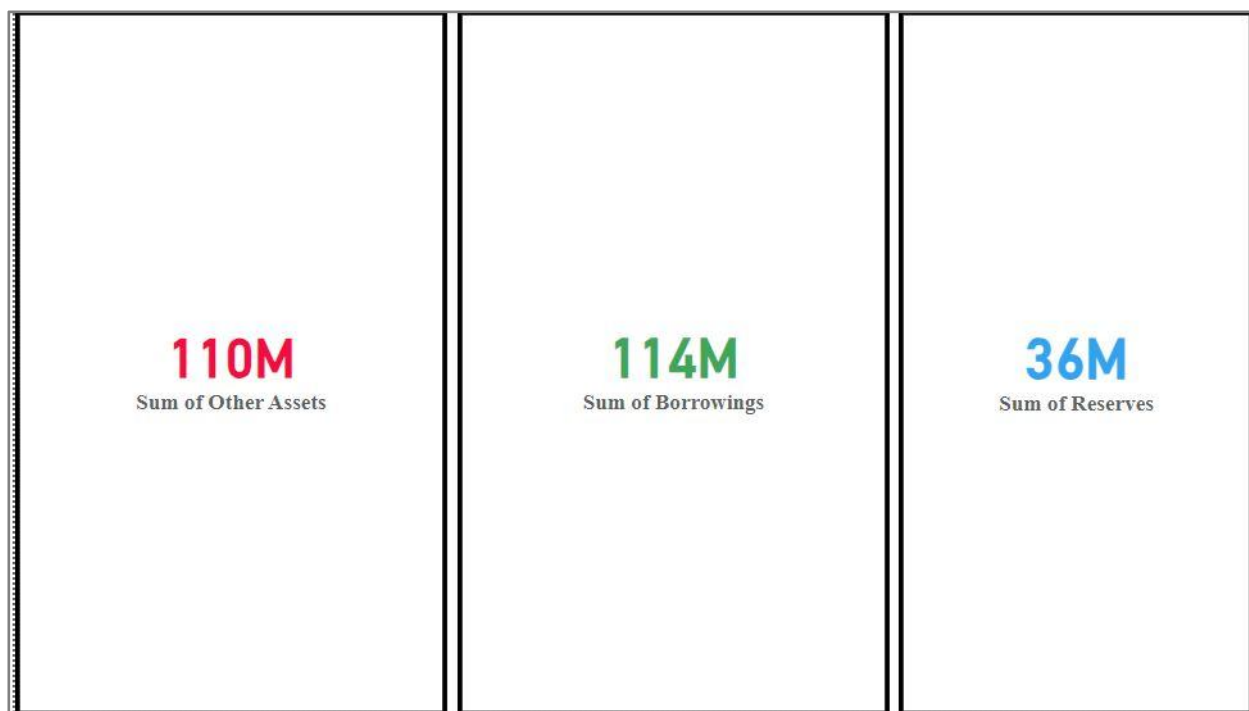
- Fields section of the Card - Total Assets
- The card will display the Total Assets value.
- Helps understand the overall financial strength of the company.
- A rising trend indicates business expansion.

Borrowings

- Fields section - Borrowings
- This will show the total borrowings amount.
- Shows the total liabilities in the form of loans or credit.
- If borrowings are too high, it may indicate financial risk.

Reserves

- Fields section - Reserves
- This will display the total reserves value.
- Represents the company's retained earnings or financial buffer.
- Higher reserves mean strong financial stability for future investments.



(Figure 3.8 KPI Card)

3.2 SKILLS MASTERED

I gained data visualization, financial statement and building dashboard competencies through the training in 30-hour Building Dynamic Financial Dashboard using Power BI. Training helped me master ways in building dashboards, conducting financial Assets and Investments. All these competencies are necessary in determining trends in business, enabling better financial planning, and making informed decisions regarding data.

3.2.1 Designing Advanced and Appealing Dashboards in Power BI

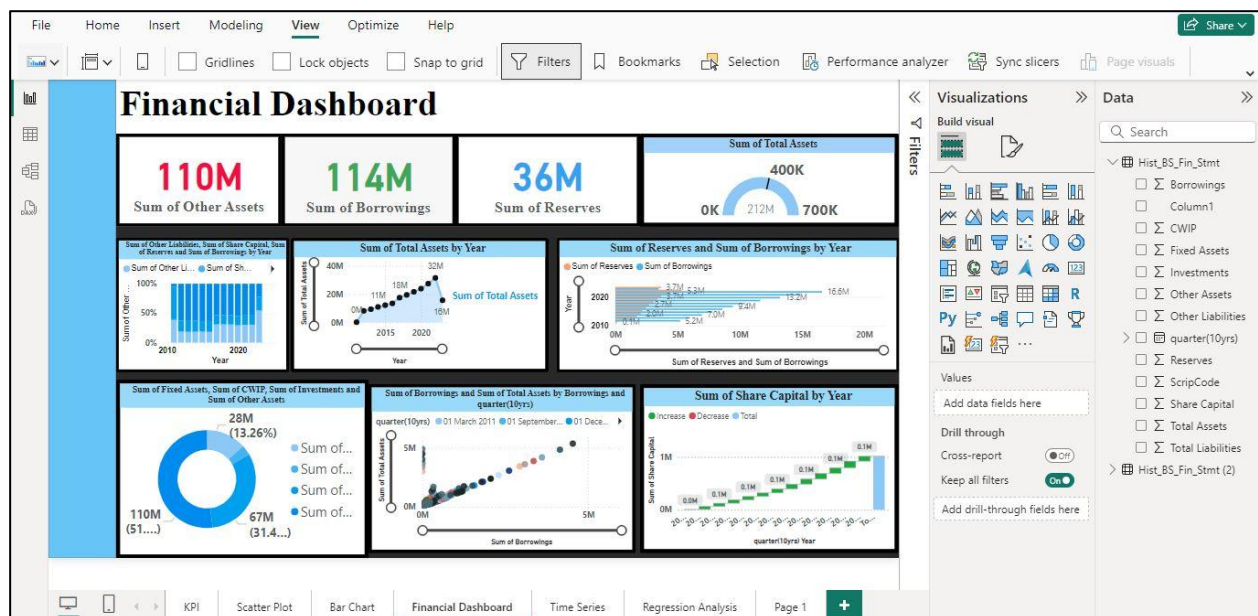
- The most notable achievement I gained was in developing dynamic and attractive Power BI dashboards. A well-crafted dashboard allows users to engage with data, analyze trends, and inform business decisions.

Through this training, I learned how to

- Make use of diverse visualization alternatives including Bar Charts, Line Charts, Pie Charts, Gauge Chart, and KPI Cards to dashboards.

- Make themes, colors, and styling to present dashboards in an attractive and readable manner.
- Utilize drill-throughs, slicers, and filters to support dynamic user interactivity with multiple data points.
- Unify diverse datasets in diverse sources (cloud environments, SQL databases, Excel spreadsheets) to craft an enterprise-level business dashboard.
- Automate data updates and schedule refreshes to maintain dashboards in synchronization with the most current data.
- Enhance dashboard performance through reducing data load time while ensuring smooth interaction.

I can now create interactive, live, and dynamic dashboards to support companies in performance tracking and making informed decisions by excelling in dashboard development.



(Figure 3.9 Power BI Dashboard)

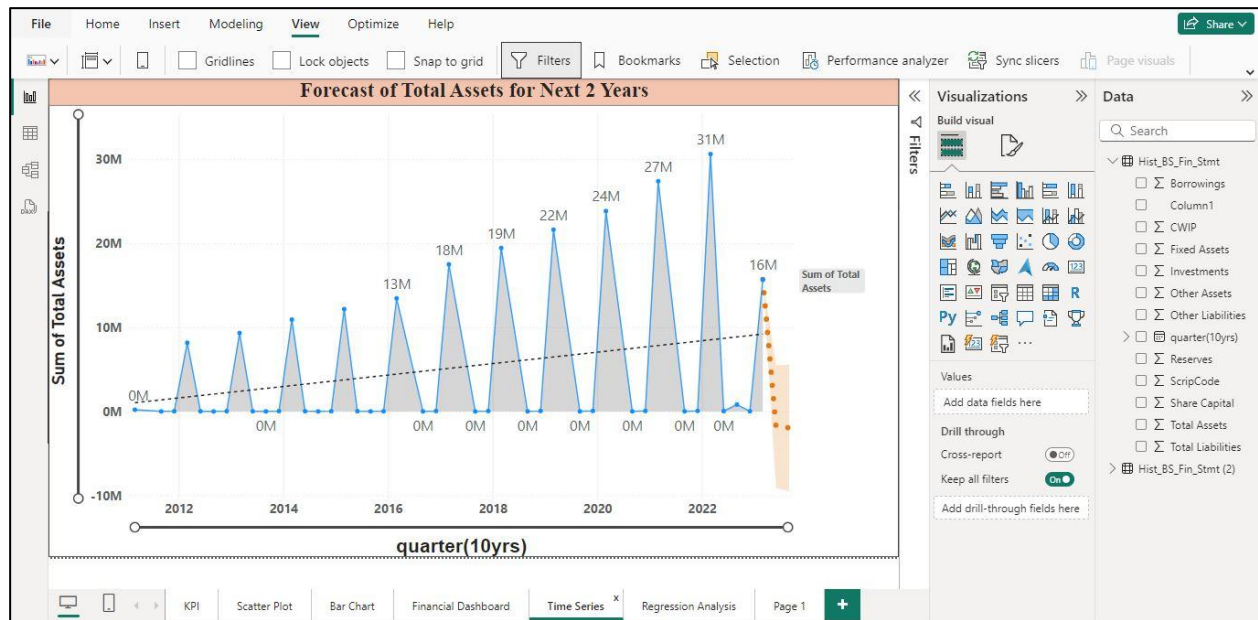
3.2.2. Performing Time Series Analysis to Obtain Business Intelligence

Time series analysis is one of the most powerful tools to work with changes in assess financial stability, investment strategies, and overall business growth. The line chart visually represents

the changes in Total Assets over multiple quarters. This training provided experience in analyzing finance-related data by working through time series models.

I learned –

- This chart represents the historical trend and future forecast of Total Assets (average across companies) over time.
- The peaks and troughs suggest a cyclical pattern, where assets increase and then drop periodically.
- The forecasted trend shows a steady increase in Total Assets over the next 8 quarters.
- Unlike the historical data, the forecasted values appear smoother, as statistical models reduce extreme fluctuations.
- This suggests that the model expects gradual growth, assuming no major external shocks.
- The historical fluctuations indicate that external factors (economic conditions, company strategies, or financial policies) have influenced asset trends.
- The forecast suggests a positive outlook, but it may not fully capture short-term shocks or volatility.
- Further analysis needed: If the data shows seasonality (regular fluctuations), a more advanced model like ARIMA or Prophet could improve accuracy. I can now support companies in examining changes in financial and identifying growth potential through mastery in time series analysis.



(Figure 3.10 Time Series Analysis Line Chart)

3.2.3 Using Regression Analysis in Power BI to Predict Investments based on Total Assets.

Before performing regression, I have verified if Total Assets has a strong correlation with Investments. The goal of the line chart is to use regression analysis to forecast future values of Investments or Liabilities based on Total Assets. This helps us analyze trends, relationships, and potential future growth.

Regression Analysis helps us -

- Identify relationships between variables.
- Forecast future trends based on historical data.
- Make data-driven decisions for financial planning.
- X-Axis – Total Assets
- Y-Axis - Investments
- Power BI predicts future values of Investments based on past trends.
- It plots a solid forecasted line on the graph.
- The shaded area represents confidence intervals (uncertainty in prediction).
- This red line represents the best-fit line for the given data.

- It shows the overall trend of the relationship.
- The line moves upward, it suggests that higher Total Assets lead to increased Investments.
- The line is flat, it indicates no strong relationship between the variables.
- The Solid line represents predicted future values based on past data.
- It extends beyond the last data point to project future trends.
- This helps in decision-making and financial planning.

Upward Sloping Line

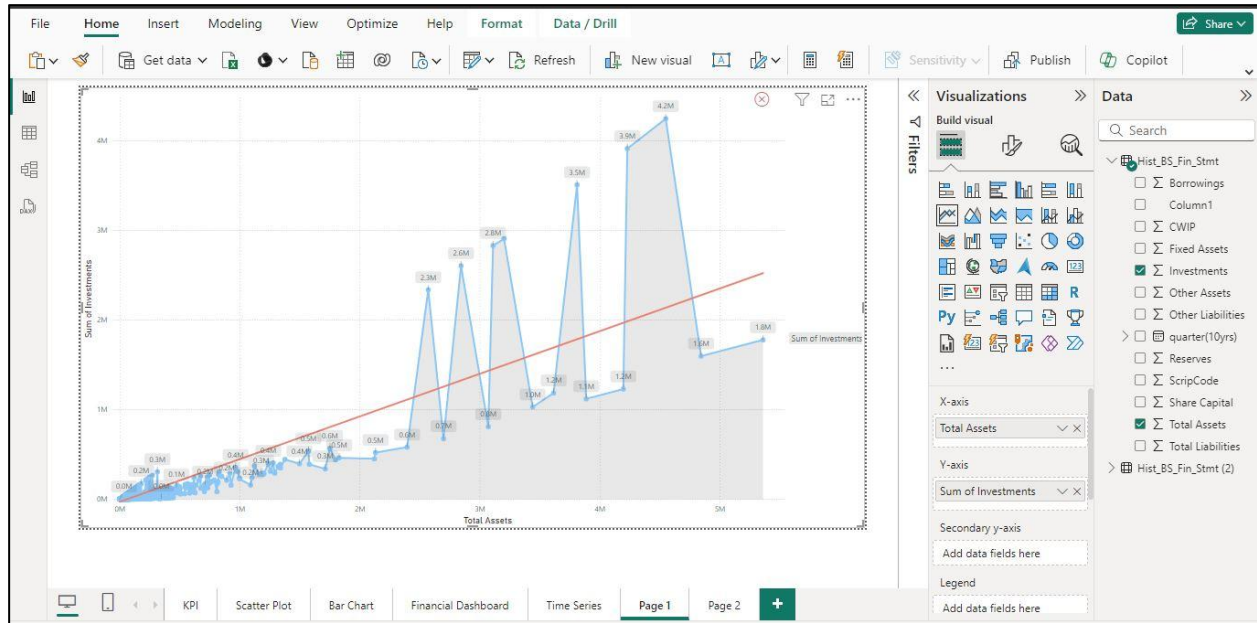
- If the trend line is going up, it indicates a positive correlation.
- Example: As Total Assets increase, Investments also increase.
- This suggests a healthy financial growth pattern.

Downward Sloping Line

- If the trend line goes down, it suggests a negative correlation.
- Example: As Total Assets increase, investment decrease (which may indicate better financial stability).
- This is a good sign for reducing debts over time.

Flat or No Slope

- If the line is flat, it means there is no significant relationship.
- Example: Total Assets may not directly impact Investments



(Figure 3.11 Regression Analysis Line Chart)

3.3 PROOF FOR OPERATIONS USING SKILLS

3.3.1 Step 1: Get Data from Sources

Financial Dashboard forecasting is started through the gathering of information from various sources such as CRM systems, channels of sales, finance databases, and marketing tools. This information helps to create an elaborate customer profile through the analysis of behavior like purchasing behavior, website visitations, opening emails, and advertising reactions. With this pattern analysis and comparison to historical successful transactions, businesses are able to accurately project future financial statements.

With Power BI, integrating the data is easily done through clicking the "Get Data" button and choosing connectors to sources such as Excel, CSV, SQL Server, Salesforce, or Google Analytics. This provides real-time tracking of the financial, analysis of trends, and evidence-based decision making to optimize profitability.

Navigator

Display Options

- Financial Dataset.xlsx [1]
 - ☒ Hist_BS_Fin_Stmt

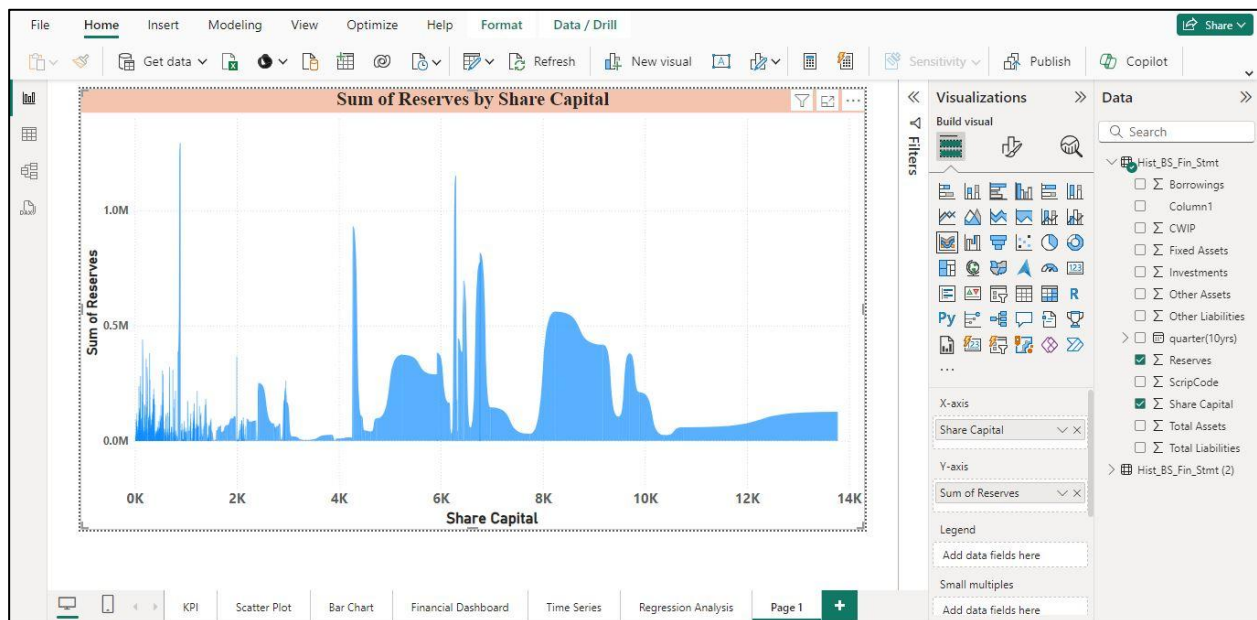
Hist_BS_Fin_Stmt
Preview downloaded on 19 March 2025

Column1	Share Capital	Reserves	Borrowings	Other Liabilities	Total Liabilities	Fixed Assets	CWIP	Investments	Other Assets	T
0	2979	166467	92447	65298	327191	138814	25363	38596	12441	
1	2936	179119	107219	73083	362357	133487	49952	42848	13607	
2	2940	195747	138761	91395	428843	141417	91494	60602	13533	
3	2943	215556	168251	117736	504486	156458	166462	76451	10511	
4	2948	228608	194714	172727	598997	184910	228697	84015	10137	
5	2959	260750	217475	225618	706802	198526	324837	82899	10054	
6	5922	287584	239843	277924	811279	403885	187022	82862	13750	
7	5926	381186	307714	302804	997630	398374	179463	235635	18415	
8	6339	442827	355133	358716	1169015	532658	109106	276767	24448	
9	6445	693727	278962	340931	1320065	541258	125953	364828	28802	
10	6765	772720	319158	399979	1498622	627798	172506	394264	30405	
11	6766	814387	335134	557219	1713506	899786	117259	336633	35982	
12	296	29284	127	11551	41157	8662	1446	1350	2969	
13	296	38350	232	13154	51932	9828	1895	1897	3831	
14	196	48999	297	17337	66829	9544	3168	3434	5068	
15	196	50439	358	22325	73318	11638	2766	1662	5725	
16	197	70875	245	16974	88291	11774	1670	22822	5202	
17	197	86017	289	15830	102333	11701	1541	41980	4711	
18	191	84937	247	19751	105126	11973	1278	36008	5586	
19	375	89071	62	24393	113901	12290	963	29330	7131	
20	375	83751	8174	27820	120120	20928	906	26356	7193	
21	370	86063	7795	35764	129992	21021	926	29373	7867	
22	366	88773	7818	43967	140924	21298	1205	30485	8793	

Load Transform Data Cancel

(Figure 3.12 Data Set)

3.3.2 Step 2: Create a Ribbon Chart



(Figure 3.13 Ribbon Chart)

The image shows a ribbon chart created in Microsoft Power BI with financial data. The chart represents the Sum of Reserves on the Y-axis against Share Capital on the X-axis. A ribbon chart

is used to show rankings over time or across different categories. However, in this case, the chart appears more like an area chart, possibly due to the dataset's structure. From the dataset in the Power BI interface, the following fields are used -X-axis: Share Capital & Y-axis: Sum of Reserves. Fluctuations in Reserves: The sum of reserves shows a lot of variations as share capital increases, indicating different financial conditions for companies at different levels of share capital. Spikes and Peaks: There are notable peaks at certain points (e.g., around 2000, 6000, and 8000 share capital), which may indicate companies with large reserves despite relatively small share capital. Steady Trend: After 10000 in share capital, the reserves appear to stabilize, showing less variation.

CHAPTER - 05

LEARNING EXPERIENCE

The 30-hour offline training in Building Dynamic Financial Dashboards Using Power BI was a learning experience rich in improving technical, analytical, and problem-solving skills. The training stressed learning about trends in Finance, application of time series forecasting methodology, and designing interactive dashboards for businesses for making informed financial decisions. The training consisted of real-time datasets, application of forecasting models, and designing industry-level dashboards in Power BI. The interweaving of theoretical and practical sessions made this training highly efficient and fruitful.

5.1 BRIEF ON NEW SKILLS LEARNED

During this training, I have gained experience in business intelligence, forecasting for Finance analysis, and visualization in Power BI. The key skills gained are –

- **Time Series Forecasting-** Time series forecasting involves predicting future values based on past data. Methods includes moving averages, trend line, and ARIMA to offer accurate projections.
- **Advanced Data Visualization** – Creating bar graphs, column graphs, donut chart, ribbon graphs, tree maps, waterfall graphs, line graphs, and decomposition trees for easy and interactive presentation of financial dashboard insights.
- **Dashboard Development** – Mastering interactive and engaging Power BI dashboards for financial decision-making and financial reporting.
- **Data Processing in Power BI** – Processing and shaping data in Power Query and DAX (Data Analysis Expressions) for efficient cleaning and transformation and preparation for accurate analysis.

They have provided me with a sound foundation in financial analytics and business intelligence and prepared me for real application in financial, retail, healthcare, and consultancy careers.

5.2 CHALLENGES FACED DURING TRAINING

Like in all learning processes, there were some difficulties in designing forecasting models and dashboards. Some of them are:

5.2.1 Understanding Complex Forecasting Models

Challenge: Statistical modeling processes such as ARIMA and exponential smoothing needed comprehensive knowledge of mathematical theory.

Solution: I managed to get over this by practicing on various datasets and reading actual life cases in order to understand how these models behave in different circumstances.

5.2.2 Handling Big Data and Optimizing for Performance

Challenge: Struggling to cope with vast amounts of information in Power BI generated slow processing and report times.

Solution: I have become proficient in optimizing data processing in Power Query and applying DAX formulas for better efficiency.

5.2.3 Creating Interactive and Usable Dashboards

Challenge: Easiest to do at early stages was to make dashboards easy to use and aesthetically pleasing in character.

Solution: I emphasized best practices in dashboard design, such as selection of appropriate charts, logical organization, and use of interactive filters.

5.2.4 Forecast Accuracy and Model Adjustments

Challenge: Challenges were experienced since certain forecasts did not surpass actual total assets amounts as a consequence of external market forces.

Solution: Enhancement in Forecasting Accuracy I tuned parameters within the model and included variables outside the model within the analysis.

Overcoming them increased confidence in me for handling forecasting projects and making recommendations using data for companies.

CONCLUSION

This 30-hour Power BI training experience on Building Dynamic Financial Dashboard using Power BI was truly extraordinary in uniting theory and application. It offered me a sound understanding of financial dashboard, business intelligence, and data visualization and equipped me with key competencies for data-informed decision-making.

I am familiar with looking at past financial trends, employed advanced tools to forecasting techniques like moving averages, time series analysis, and ARIMA modeling, and developed interactive dashboards using Power BI. They play a crucial role in determining market trends, financial plan optimization, and business decision-making.

In spite of obstacles like mastering complicated statistical modeling, handling huge databases, and creating user-friendly dashboards, I overcome them through practicing, troubleshooting, and optimization methods based on Power Query and DAX. In addition to improving the quality of my technical ability, these experiences also developed my thinking and problem-solving abilities.

This training has helped me prepare to be a financial analyst, strategic planner, and business intelligence specialist. Because organizations are increasingly focusing on information-based decision-making, this skill and expertise gained in this program will serve as a solid base for real-time problem-solving and making significant contributions toward business success.